Sean Bowman

Melbourne, FL | (XXX) XXX-XXXX | seanbowman9@gmail.com | LinkedIn | GitHub

SUMMARY

Director of Propulsion leading the engineering organization behind Vaya Space's Dauntless supercritical oxygen expander cycle hybrid rocket engine. Owns the full system lifecycle: design, analysis, manufacturing, test operations, and flight qualification. Drives a department that integrates program management and mission assurance into its engineering scope, owning schedule, risk, and deliverables across the program. Background spans hybrid engine development, thermo-fluid analysis, and engineering software architecture, with hands-on responsibility from conceptual design through control-room test execution and post-test validation.

EXPERIENCE

Vaya Space — **Cocoa, FL**

Director of Propulsion — *January 2026 – Present*

Chief Engineering Council: one of four department directors (GNC, Avionics, Structures, Propulsion) collectively holding company-wide Chief Engineer authority; sets engineering standards, technical levels, and program priorities atop full Director-level departmental scope.

- Lead the propulsion department accountable for all deliverables on the Dauntless hybrid engine, a supercritical oxygen expander cycle with swirl injection, across the full system lifecycle: design, analysis, manufacturing, test operations, and flight qualification
- Direct a team of 8 engineers and developers; own departmental schedule, personnel, and budget, and carry program management and mission assurance for the propulsion program
- Architected the 38,000+-line Python propulsion design suite with 100+ documentation files; established coding standards, onboarding, and a main/dev/feature Git workflow with mandatory reviews and quarterly releases
- Designed multi-user frontend infrastructure for concurrent engineering operation: priority-based CPU allocation, background job processing with live progress streaming, and session management with automatic resource cleanup

Aerospace Engineer II – Fluid Thermal Control — *April 2024 – January 2026*

- Took full product ownership of the regeneratively-cooled nozzle: end-to-end contour and cooling-channel design via Axisymmetric Method of Characteristics, 3D spirally-fluted geometry for LPBF additive manufacturing in GRCop-42, and CAD release
- Served as responsible engineer through nozzle fabrication and test: vendor and manufacturing oversight, DFM reviews, part inspection and assembly, control-room operations, and post-test validation against design predictions

Aerospace Engineer I – Fluid Thermal Control — *April 2022 – April 2024*

- Designed, analyzed, and tested the regenerative cooling architecture of the Dauntless nozzle: LOX coolant path design for supercritical oxygen operation and conjugate heat transfer analysis with REFPROP/CEA integration
- Originated the propulsion design suite, authoring the initial MATLAB implementation and later porting it to an object-based Python library to cut licensing cost and broaden access; expanded it from a focused regenerative-cooling toolset to the full engine design environment

Graduate Research Assistant — **Florida Institute of Technology**

Spring 2021 – Fall 2021 — *Melbourne, FL*

- Performed DPM CFD simulation of cryogenic droplet thermo-fluid mixing; achieved <5% deviation from experimental benchmarks
- Delivered validated analysis tool to leading rocket manufacturer; analysis article currently in active use

Graduate Teaching Assistant Lead — **Florida Institute of Technology**

January 2019 – May 2021 — *Melbourne, FL*

- Created course materials and lectured for AEE 3064 Fluid Mechanics Laboratory; oversaw a team of instructors conducting weekly progress meetings

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, JavaScript, HTML, FORTRAN, VBS, Git

Engineering Tools: Siemens NX, Teamcenter (PLM), Simcenter Nastran, STAR-CCM+, ANSYS Fluent, SolidWorks, Atlassian Suite (Jira, Confluence, Bitbucket), LaTeX, REFPROP, CEA, GFSSP

Process: Test Planning & Operations, Risk Assessment, Program Management, Mission Assurance, Design for Manufacturability & Assembly, Additive Manufacturing, Code Review, Technical Documentation

Leadership: Team Direction & Mentorship, Standups & 1:1s, Executive & Board Engagement, Resource & Budget Planning, Engineering Standards & Infrastructure, Investor Relations, DoD / NASA Interface

EDUCATION

Master of Science in Aerospace Engineering — December 2021

Florida Institute of Technology — Melbourne, FL

Bachelor of Science in Applied Physics, Minor: Mathematics — December 2017

Stockton University, Magna Cum Laude — Galloway, NJ